

Step: 1: Hashing the documents,

Given, Hash function: $H(doc) = (char\ count \times id \times 100) \% 360$

Document	ID	Char Counts	Hash (doc)
lastname.txt	1	12	$(12 \times 1 \times 100) \% 360 = 120$
yourid.txt	2	10	$(10 \times 2 \times 100) \% 360 = 200$
Hello.mp4	3	9	$(9 \times 3 \times 100) \% 360 = 180$
World.txt	4	9	$(9 \times 4 \times 100) \% 360 = 0$
Again- class.mp4	5	15	$(15 \times 5 \times 100) \% 360 = 300$
final-quiz-firstname.txt	6	24	$(24 \times 6 \times 100) \% 360 = 0$

[Note: characters count include everything e.name and extension.]

Step: 02:- Hashing the server.

$$H(x) = \text{Server id} * 90 \% 360$$

Server	ID	Hash
A	1	$1 \times 90 \% 360 = 90$
B	2	$2 \times 90 \% 360 = 180$
C	3	$3 \times 90 \% 360 = 270$
D	4	$4 \times 90 \% 360 = 0$

Step: 03:- partitioning the servers,

Given, the weight of all server is 10,

And total partitions is 15.

So, partitions per server

$$= \frac{\text{Server weight}}{\text{Total weight}} \times \text{Total partition}$$

$$= \frac{10}{40} \times 15$$

$$= 3.75 \approx 3-4$$

P.T.O

For, Server A (Hash = 90)

$$A0 = 90^\circ$$

$$A1 = (90 + 1 * 10) \% 360 = 100^\circ$$

$$A2 = (90 + 2 * 10) \% 360 = 110^\circ$$

$$A3 = (90 + 3 * 10) \% 360 = 120^\circ$$

For, Server B (Hash = 180)

$$B0 = 180^\circ$$

$$B1 = (180 + 1 * 10) \% 360 = 190^\circ$$

$$B2 = (180 + 2 * 10) \% 360 = 200^\circ$$

$$B3 = (180 + 3 * 10) \% 360 = 210^\circ$$

Similarly,

$$C0 = 270^\circ$$

$$C1 = 280^\circ$$

$$C2 = 290^\circ$$

$$C3 = 300^\circ$$

and

$$D0 = 0^\circ$$

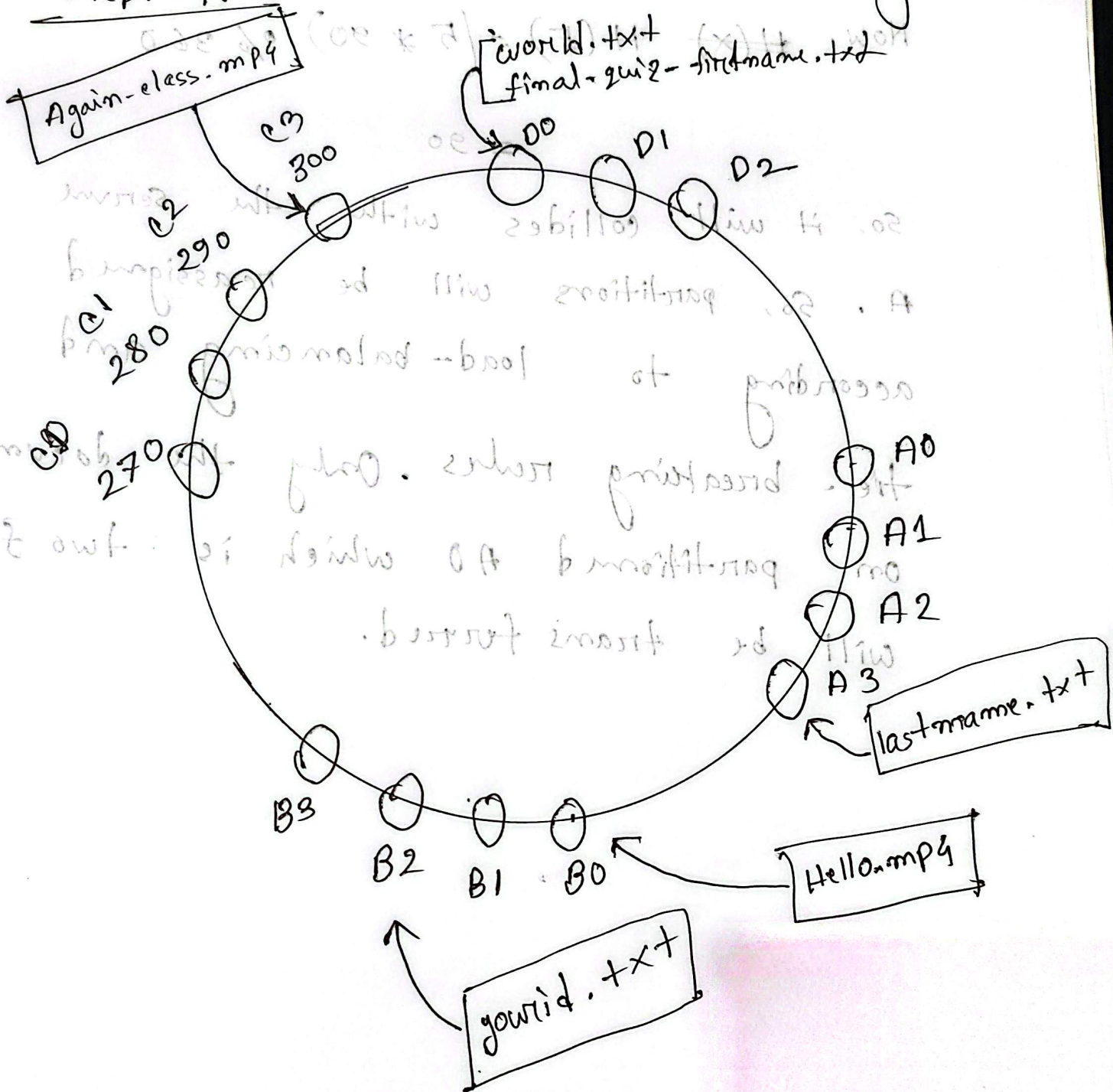
$$D1 = 10^\circ$$

$$D2 = 20^\circ$$

P.T.O

Step: 04:

Files on the ring :-



Now, $H(x) = H(E) = (5 * 90) \% 360$

So, it will collide with the server
A. So, partitions will be reassigned
according to load-balancing and
tie-breaking rules. Only the documents
on partitioned A0 which is two files
will be transferred.

